

Day 3: RAM and Storage

Easy-to-Learn Theory + Practical Assignment Answers

1. DDR3 vs DDR4 vs DDR5 RAM

Feature	DDR3	DDR4	DDR5
Generation	Older	Newer	Newest of these three
Typical Speed	800–2133 MT/s	1600–3200 MT/s	Starts around 4800 MT/s and can go much higher
Voltage	1.5V	1.2V	1.1V
Performance	Lower	Good	Higher
Typical Use	Older PCs/laptops	Many PCs/laptops	Newer PCs/laptops
Power Use	Higher	Lower	Lower
Gaming	Basic/older systems	Good	Better for newer systems

Easy Memory: DDR3 → Old + Slower | DDR4 → Common + Good | DDR5 → New + Faster

Important: DDR3, DDR4 and DDR5 are not interchangeable. The motherboard must support the specific RAM generation.

2. RAM Removal and Reinstallation

Safety: Only open a computer you are allowed to open. If you are unsure, do not force the RAM or open the device.

1. Shut down the computer completely.
2. Disconnect the charger or power cable.
3. Open the computer according to the manufacturer's instructions.
4. Find the RAM slot and RAM module.
5. Take a clear photo of the installed RAM module for the assignment.
6. Release the retaining clips and gently remove the RAM.
7. Align the notch on the RAM with the slot.
8. Press the RAM gently and evenly until the clips lock.
9. Close the computer and reconnect the power.
10. Turn it on and check that the computer boots normally.

Assignment photos: ■ Photo 1 — RAM installed in the slot. ■ Photo 2 — RAM after reinstalling, if required.

3. HDD vs SSD vs NVMe

Feature	HDD	SSD	NVMe SSD
Technology	Magnetic disk	Flash memory	Flash memory using PCIe
Speed	Slowest	Faster	Usually fastest
Moving Parts	Yes	No	No
Shock Resistance	Lower	Higher	Higher
Connection	Usually SATA	SATA or other interfaces	Usually M.2 + PCIe
Typical Use	Large, inexpensive storage	OS, apps and games	Fast OS, apps and games
Daily Use Example	Storing photos/videos	Windows and everyday apps	Demanding games/large apps

Easy Memory: HDD = Slow + Cheap + Large | SSD = Fast + Silent | NVMe = Very Fast + PCIe

Important: M.2 describes a physical form factor/slot. NVMe describes a storage protocol. Not every M.2 drive is NVMe.

4. Storage Drive Removal and Reinstallation

Safety: Only perform this task on a device you are allowed to open and only if you know the correct procedure for that model.

1. Turn off the computer completely.
2. Disconnect the charger or power cable.
3. Open the device only according to the manufacturer's service instructions.
4. Find the storage drive: it may be a 2.5-inch SATA HDD/SSD or an M.2 SSD.
5. Take a clear photo showing the connector type.
6. Carefully remove/disconnect the drive following the manufacturer's instructions.
7. Put the drive back in the correct position and reconnect it securely.
8. Close the device before reconnecting power.
9. Turn on the computer and check that the operating system recognizes the storage drive.

■ Connector Types

SATA DRIVE

Computer ■■■■■ SATA ■■■■■ HDD / SATA SSD

M.2 / NVMe SSD

Motherboard ■■■ M.2 Slot ■■■■ NVMe SSD

■ Day 3 Quick Revision

RAM → Temporary memory used while programs are running.

DDR3 → Old / slower generation

DDR4 → Common / good performance

DDR5 → Newer / faster generation

HDD → Slow + large + usually cheaper

SSD → Fast + silent + no moving parts

NVMe → Very fast PCIe-based storage